

AKSHAT RAJ

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SUMMARY

Machine Learning Developer with hands-on experience building end-to-end predictive models, REST APIs, computer vision systems, and LLM-powered multi-agent applications. Skilled in Python, Scikit-learn, TensorFlow, FastAPI, and LangChain, with a strong foundation in data preprocessing, feature engineering, model training, evaluation, and deployment.

EDUCATION

Vellore Institute of Technology, Bhopal

Expected May 2026

B.Tech in Computer Science in Cloud Computing — CGPA: 8.12

PROJECTS

Mental Health Score Prediction / Python, Scikit-learn, FastAPI, Pandas, HTML/CSS/JavaScript

July 2026

- Built an end-to-end regression pipeline predicting a 0–10 student mental health score from academic, lifestyle, and social media usage data, using a Scikit-learn ColumnTransformer to apply log-transformation, scaling, ordinal, and one-hot encoding across mixed feature types.
- Trained and benchmarked Linear Regression and Random Forest Regressor models, tuning hyperparameters with RandomizedSearchCV (5-fold cross-validation, 15 iterations); the Random Forest model achieved 87.8% R^2 and 0.35 MAE on held-out test data.
- Deployed the trained model as a REST API using FastAPI with Pydantic schema validation and CORS support, and built a responsive HTML/CSS/JavaScript frontend with real-time client-side validation, an animated score gauge, and graceful error handling for live predictions.

Multi-Agent AI Research Assistant / Python, LangChain, Groq (Llama-3.3-70B), Streamlit

June 2025

- Architected a 4-agent research pipeline (Search, Reader, Writer, Critic) using LangChain and Groq's Llama-3.3-70B to autonomously research a topic and generate structured, source-cited reports.
- Built custom LangChain tools integrating the Tavily Search API and BeautifulSoup-based web scraping to retrieve and parse real-time content from top search results.
- Implemented a critic agent that scores generated reports and returns structured feedback (strengths, gaps, verdict), enabling automated quality review without human input.
- Deployed an interactive Streamlit app with real-time pipeline status tracking, tabbed report/critique/raw-research views, and one-click Markdown export.

Brain Tumor Detection via MRI Classification / Python, TensorFlow, OpenCV

March 2025

- Developed a CNN classifier for brain MRI scans across 4 categories (Glioma, Meningioma, Pituitary, No Tumor), achieving 93.18% test accuracy.
- Engineered an automated preprocessing pipeline using OpenCV contour detection to isolate brain tissue, reducing background noise and standardizing 512×512 inputs.
- Implemented Keras ImageDataGenerator augmentation and Early Stopping + Model Checkpointing callbacks; built an interactive Streamlit diagnostic interface for instant classification with confidence scores.

Real-time Sign Language Recognition System / Python, TensorFlow, OpenCV, MediaPipe

April 2024

- Designed and trained a custom LSTM neural network on 5,200+ images across 26 gesture classes, achieving 96–100% classification accuracy on real-time webcam input.
- Built a real-time data pipeline using OpenCV and MediaPipe to track 21 hand keypoints per frame at 30+ FPS, enabling sub-100ms end-to-end inference latency.
- Deployed an interactive web demo using Streamlit with on-screen confidence score visualization and data augmentation (flip, rotation, brightness jitter) for improved generalization.

TECHNICAL SKILLS

Programming Languages: Python

Database: MySQL, SQL

AI / LLM Frameworks: LangChain, Groq API, Multi-Agent Systems, RAG

Machine Learning: TensorFlow, Keras, Scikit-learn, CNN, LSTM, Random Forest, Linear Regression, Hyperparameter Tuning

Computer Vision: OpenCV, MediaPipe

Data Analysis: Pandas, NumPy, Matplotlib, Seaborn

Tools & Platforms: Git, GitHub, Streamlit, Jupyter Notebooks

CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification – DeepLearning.AI
- Intro to Machine Learning – Kaggle
- Introduction to Artificial Intelligence – Infosys